Same Parts, Different Arrangements Matched Edits for Budgeted Anomaly Inspection

Supplementary Material

1. Inspection protocol

This supplement provides the decision rules, training details and complete results underlying the main paper. The inspection protocol, comprising the cutoffs, the endpoints and the decision criteria, was specified before any test score was examined. The upstream-only continuation (Sec. 3) was added before test scoring, after development data showed that the class-wide rule-aware heads did not meet the compartment-ranking criterion, and the eligibility rules and edits of the juice experiment were fixed before any juice photograph was scored. All policies, seeds and interfaces are reported.

Scene and compartment cutoffs. The scene cutoff is the 63rd smallest maximum among 69 normal validation maps, and an alarm requires a strictly greater score. The compartment cutoff at recall r is the $\lceil 138r \rceil$-th largest development target score among scenes that raise an alarm; both the vacated and the recipient compartment are included, even when one of them is difficult to localize, and the comparison is inclusive. If fewer than $\lceil 138r \rceil$ targets remain after the scene alarm, the cutoff is set to zero and the operating point is reported as infeasible. The fixed-cost cutoff is the $(\lfloor 0.5\, n_{\mathrm{dev}} \rfloor + 1)$-th largest score among the untouched compartments of development faults that raise an alarm, with a strict comparison, which admits at most 0.5 untouched compartments per development fault; it is likewise set to zero if too few negatives remain. Any compartment queued on a source or a sham counts as a false mark, and for the vacancy-only variant the vacated compartment is the only positive. Rank statistics are computed on the raw maps before the scene alarm, with exact ties broken by compartment index.

Decision criteria. The primary endpoint is the difference in untouched compartments queued per extracted fault between the footprint and the class-wide policy. A difference is considered useful if it removes at least one compartment with a 95% interval that excludes zero, while the achieved target recall remains at least 90% and declines by at most five points, the fault and vacancy alarm rates remain at least 95% and decline by at most three points, and valid-sham alarms increase by at most five points. A mechanistic interpretation further requires that the class-wide control reach a development alarm rate of at least 90% for both fault types, a fault-over-sham win rate of at least 90% and a compartment AUROC of at least 90%. Explaining the behaviour of the released model additionally requires that the class-wide control produce broad changes in untouched compartments, exceeding those of the vacancy and the sham. The seed replicates and the two photo halves are required to preserve the sign of the primary contrast. All intervals are conditional on the fitted heads and cutoffs.

Region area and diagnostic response. The region area uses binary saturation: each compartment is counted once, when its score crosses a common threshold. Positives are the two target compartments of each fault; negatives are its thirteen untouched compartments together with the 30 compartments of the corresponding source and sham, or only the 13 untouched compartments for the fault-only variant. Recall is interpolated linearly through ties and integrated up to a pooled false-compartment rate of 0.05, and the area is divided by 0.05. Both curves are descriptive sweeps over the test set, whereas the scene and compartment cutoffs are fixed on development data; bootstrap replicates resample complete photo groups and recompute the pooled sweep. In contrast to the official pixel-level AU-sPRO, which allows area-based saturation inside ambiguous native regions [1], our compartments carry an exact label and constitute indivisible units of inspection.

The any-pixel summary counts a compartment as changed whether it contains a single large change or a broad one. The complementary area fraction counts pixels whose absolute change is at least 0.5 on the same 13 compartments, weighted by compartment area within each original photograph, and we additionally report per-compartment upper quantiles averaged over compartments. These comparisons with the source serve diagnosis only and do not enter the inspection decision.

2. Paired fine-tuning

Target construction. We build on the released SALAD code (revision `753ba94`) and checkpoints [2]. When a part is inserted, the function `copy_label_diff_img` writes the donor footprint into the edited six-channel map and returns the entire resulting class plane as the discriminative target; `copy_and_change` takes the union of this insertion mask and the same-image relabelling mask. The reconstruction objective uses the clean source labels with multiclass focal and Dice losses and is unaffected by these targets. The discriminative objective is

$$5\,\mathrm{mean}_p\,\mathrm{FL}_{\alpha,\gamma}(z_p, y_p) + \mathrm{mean}_p\,|\sigma(z_p) - y_p|,$$

with the default sigmoid focal-loss parameters $\alpha = 0.25$ and $\gamma = 2$. Both terms average uniformly over all pixels of a batch, without area-dependent reweighting. The footprint policy records the paste mask before it is expanded to the class plane and modifies only the insertion part of the target, leaving the relabelling target of combined operations intact. Both policies share the Perlin perturbation, the bilinear rotation and the donor insertion and relabelling functions of the released code. Since the autoencoder is frozen, the reconstruction objective does not update either U-Net.

Training schedule. Training uses 372 normal training photographs and 371 eligible rule-aware training edits. Each batch of four contains two clean maps, one SALAD-style synthetic edit and one rule-aware edit, drawn as a relocation, a sham or a vacancy with probabilities $1/2$, $1/4$ and $1/4$; the four SALAD edit families are sampled uniformly. In the upstream-only continuation, the rule-aware slot is replaced by its clean source while all other slots, donors and transformations are kept, which preserves the 3:1 ratio of clean to synthetic examples of the original loader. We use Adam with weight decay 10^{-5} and an initial learning rate of 3×10^{-5} for rule-aware fine-tuning and 10^{-5} for the continuation, train for 800 steps with batches of four, and decay the learning rate by a factor of 10 after step 760. The final checkpoints of seeds 3407 and 3419 (seeds 1 and 2 in the main paper) are evaluated. Within each pair of runs, the initial checkpoint and the sequence of input batches are identical and only the targets differ, which we verified for every example. At inference, the fine-tuned head replaces the released head of identical architecture.

3. Continuation of the original recipe

The rule-aware class-wide heads detect all faults and vacancies but fail the localization criterion. We therefore evaluate a separately specified continuation in which the augmented slot contains only the original synthetic edit families, with the released head as the reference. A paired effect in this setting identifies the influence of the target policy during continued training. A full counterfactual of the released model would additionally retrain the reconstruction network from the start of the original 70,000-step schedule. Since the two policies mark different numbers of pixels, the positive area remains part of the intervention. A separately locked continuation uses the original synthetic edit families alone. Its class-wide seed 2 passes development competence (97.3% extracted compartment AUROC), ranks targets at 98.1% test AUROC and produces a large change in 88.8% of untouched compartments. In that pair, footprint supervision reduces the fraction to 1.3% and the queue from 11.50 to 7.07, a paired reduction of 4.43 marks [3.72, 5.14] at 96.0% and 95.3% target recall. Both heads detect all faults and vacancies, with 18 and 20 sham alarms. Seed 1 gives a similar queue reduction and misses the development AUROC floor. This isolates an effect of insertion targets during a continuation that reproduces the broad response in one competent control. The fine-tune with rule-aware examples achieves the lowest queues.

4. Full SALAD and the liquid–label rule

Branch implementation. The released teacher, student, RGB autoencoder, map autoencoder and map U-Net are loaded from the same released archive. The RGB branch uses the released PIL resize and ImageNet normalization and uses only RGB in its local spatial map. The composition branch uses one-hot maps and a reconstruction. The Gaussian branch pools the first 384 channels of the RGB student over the full image and over semantic regions supplied by the maps. The implementation fits Ledoit–Wolf covariances on normal training features and applies validation means and standard deviations to the per-region distances. The feature statistics are refit on native normal train and validation RGB/map pairs. The official fusion sums the normalized local and map maxima and the released global score. Composition- and RGB-map spatial outputs are kept separately. We use normal-only 10% alarm cutoffs per tested branch and interface after applying the official normalizations. The extracted interface uses the same RGB and normalization statistics as the oracle route and substitutes maps from the independent extractor. Our 91 native logical and 138 good pushpins reproduce the published 100.0% full-system logical AUROC. On the matched photos the local RGB branch alarms on 128 valid shams, including 119 newly alarmed from a clear source. Both edited arms retain the same cutout and paste support, so this is evidence of an appearance response to the composited valid move. Those shams are valid under the placement rule.

Juice assets and eligibility. The dataset specifies a correspondence between the liquid and printed icon [1]. We use a fixed bank of 16 banana, 16 cherry and 16 orange donor photos from the official training split, sorted by a salted SHA-256 key among eligible cases. The upper and lower label windows are the two largest four-connected class-1 components. We place a 36×56 rectangle at the upper label's bounding-box center. At least 99.5% of the rectangle must fall within the upper-label component. Liquid type is assigned by its exposed RGB median (a dark red channel below 105 for cherry, otherwise blue at least 90 for banana or at most 75 for orange). We inspected the source, donor and final icon crops before detector scoring. The RGB intervention leaves the original label border, lower label and bottle contents fixed. It retains the original composition map. The label identity is changed within the RGB print only.

Table 1. **Positive mask area in paired rule-aware training.** Positive counts are means over examples of each edit type on a 256×256 grid. The upstream-only continuation uses the same synthetic edits and replaces the rule-aware slot with a clean map.

Seed	Input edit	Examples	Class-wide px	Footprint px
3407	Clean	1600	0.0	0.0
3407	Combined	205	28841.7	8885.4
3407	Insert	204	22838.7	1583.1
3407	Perlin	187	2670.3	2670.3
3407	Relabel	204	7852.4	7852.4
3407	Relocate	409	3822.8	460.9
3407	Vacancy	169	228.5	228.5
3407	Valid Sham	222	0.0	0.0
3419	Clean	1600	0.0	0.0
3419	Combined	204	27599.0	9901.4
3419	Insert	194	21770.8	1477.1
3419	Perlin	203	2503.0	2503.0
3419	Relabel	199	7625.5	7625.5
3419	Relocate	420	3823.8	462.4
3419	Vacancy	195	231.4	231.4
3419	Valid Sham	185	0.0	0.0

Table 2. **Rule-aware fine-tuning on oracle labels.** See the main inspection table for definitions and the extracted results. Source and sham counts are paired within originals.

Map and compartment score	Src. / Sham	Fault / Vac.	Top 2 (%)	Marks to both	Recall (%)	Untouched	Reg. area
Released, max	31 / 31	138 / 138	76.8	2.46	95.3	10.51	81.4
Released, 99th pct.	31 / 31	138 / 138	87.0	2.33	95.3	11.07	77.6
Class-wide, seed 1	16 / 24	138 / 138	1.4	14.81	96.0	13.00	39.9
Footprint, seed 1	14 / 18	138 / 138	87.7	2.12	96.0	0.20	95.4
Class-wide, seed 2	22 / 29	138 / 138	1.4	14.81	96.0	13.00	40.2
Footprint, seed 2	15 / 16	138 / 137	76.1	2.28	96.0	0.46	88.8

The two donor crops use the same vertical offset from $\{-2, 0, 2\}$ and the same horizontal offset from $\{-8, -6, -4, 0, 4, 6, 8\}$ relative to their respective label centers. They must retain the icon's dark connected strokes within the unblended core. The original recipient rectangle and taper are shared. The taper reaches full opacity three pixels from its edge. Each donor border mean absolute difference is at most 13 digital levels, with a between-arm difference at most 2.5. Mean blended RGB differences between arms differ by at most 0.5 digital levels averaged over the recipient rectangle. The pair minimizes the sum of those matching gaps. A centered-crop feasibility pilot gave systematically greater edit magnitude for faults. The offset and core checks were set without model outputs. We retained 53/54 validation and 89/94 test sources. Reused donors are fixed assets. The original test photo is the bootstrap unit. Neither same-category pushpin half is counted as a new data domain.

Targets and results. We rank the local appearance map over upper label, lower label, cap, shoulder liquid and lower liquid. Among the 89 fault maps the upper label ranks first in 66 and has mean rank 1.33. It ranks first in 76/89 shams and 5/89 originals. With fusion as the scene gate, its development-calibrated spatial cutoff recovers it in 84/89 faults and queues 1.33 other regions per fault. Local appearance gives 28 sham and 25 fault scene alarms at its own cutoff. Fusion and global appearance each detect all 89 faults and order every fault above its sham. The exact binomial 95% interval for an all-hit count is 95.9–100%. The paired bootstrap intervals for the fault-minus-sham alarm gaps are 70.8–87.6 points for fusion and 76.4–91.0 for global appearance. Composition-only yields the same output pixel for pixel across all variants, so any of its four scene alarms applies equally to each state.

5. Refitted graph readout

The mean-aggregation comparator uses the original query-masked endpoint inputs, message encoder, donor-balanced

Table 3. **Paired rule-aware differences at the pre-specified cutoffs.** Untouched compartments are per fault; target recall, sham alarms and region area are in percentage points. Each interval uses 4,000 resamples of original photographs.

Input	Footprint contrast	Untouched	Target recall	Sham alarm	Region area
Extracted	3407 vs. released	-10.88 [-11.56, -10.12]	-0.4 [-3.6, +3.3]	+1.4 [-3.6, +7.2]	+14.7 [+10.8, +18.6]
Extracted	3407 vs. class-wide	-12.80 [-12.87, -12.72]	-1.8 [-4.7, +1.1]	-7.2 [-13.8, -0.0]	+57.0 [+53.5, +60.5]
Extracted	3419 vs. released	-10.62 [-11.31, -9.86]	+0.4 [-2.9, +3.6]	+9.4 [+2.2, +16.7]	+8.0 [+3.3, +13.0]
Extracted	3419 vs. class-wide	-12.55 [-12.65, -12.45]	-0.4 [-3.6, +2.9]	-1.4 [-9.4, +6.5]	+50.3 [+46.1, +54.6]
Oracle	3407 vs. released	-10.30 [-11.05, -9.49]	+0.7 [-2.5, +4.0]	-9.4 [-15.9, -2.9]	+14.0 [+9.9, +18.1]
Oracle	3407 vs. class-wide	-12.80 [-12.86, -12.73]	+0.0 [-3.3, +2.9]	-4.3 [-11.6, +2.9]	+55.6 [+51.8, +59.2]
Oracle	3419 vs. released	-10.04 [-10.80, -9.21]	+0.7 [-2.5, +4.0]	-10.9 [-19.6, -2.9]	+7.5 [+2.5, +12.6]
Oracle	3419 vs. class-wide	-12.54 [-12.64, -12.42]	+0.0 [-3.3, +3.3]	-9.4 [-17.4, -1.4]	+48.7 [+44.3, +53.1]

Table 4. **Additional rule-aware operating points.** Top 3: both targets among the three highest compartments. The 98% and 0.5-cost cutoffs are set on development data. Recalls and the fault-only low-FPR region area are percentages. Untouched, sham and vacancy-only costs are means per image. Sham marks and vacancy-only untouched marks use the 95% development cutoff.

Input	Policy	Top 3	98 rec.	98 unt.	Cost rec.	Cost unt.	Fault region	Sham marks	Vac. unt.
Extracted	Released, max	94.2	98.9	12.56	81.9	0.30	74.0	0.00	0.39
Extracted	Released, 99th pct.	96.4	98.9	12.62	75.7	0.17	73.8	0.00	0.28
Extracted	Released, 95th pct.	92.8	98.6	12.46	73.2	0.25	73.2	0.00	0.11
Extracted	Released, reconstruction product	65.2	98.6	12.59	67.4	0.51	62.4	0.00	0.33
Extracted	Class-wide, seed 1	0.0	97.5	13.00	39.5	0.65	36.2	0.43	1.37
Extracted	Footprint, seed 1	99.3	97.5	0.22	100.0	0.38	88.4	0.01	0.14
Extracted	Class-wide, seed 2	0.0	97.8	13.00	39.1	0.64	36.0	0.09	0.59
Extracted	Footprint, seed 2	97.8	98.2	0.57	98.2	0.57	77.3	0.14	0.23
Oracle	Released, max	92.8	98.2	12.32	83.0	0.39	72.7	0.12	0.37
Oracle	Released, 99th pct.	96.4	98.9	12.60	75.4	0.25	71.9	0.11	0.28
Oracle	Released, 95th pct.	91.3	98.6	12.51	73.6	0.39	69.8	0.11	0.11
Oracle	Released, reconstruction product	69.6	98.9	12.64	71.7	0.41	67.1	0.12	0.40
Oracle	Class-wide, seed 1	1.4	98.9	13.00	38.8	0.43	36.5	0.66	1.14
Oracle	Footprint, seed 1	100.0	97.8	0.26	99.6	0.30	88.2	0.04	0.17
Oracle	Class-wide, seed 2	1.4	98.9	13.00	38.4	0.42	35.9	0.38	0.84
Oracle	Footprint, seed 2	96.4	98.9	0.64	98.2	0.58	77.9	0.13	0.23

fitting loss and learning-rate and early-stop rules. It uses mean pooling within each neighborhood and the original mean/max pooling across the endpoints. It omits explicit degree inputs and degree-scaled pooling. Its original GlaS four folds yield normal-link AUROC between 95.7% and 98.2% with better loss than a prevalence intercept in all four. All 280 query-bit flip checks give exact equality. The mean model can still use neighborhood content that covaries with degree or recognize an empty neighborhood.

The original and refitted networks use a query-excluded adjacency for all tokens. A changed observed bit can affect a loss through the scoring formula without moving the fitted probability. The pair MLP is an invariance control because geometry and roles stay fixed. The degree-only probe concentrates its changed features on the four target pairs by construction. It tests whether the fitted model responds to that input. Its perfect recovery and the fixed-weight clamp should be interpreted with the refit in Tab. 15. SpotTarget studies degree-sensitive bias from including and excluding target edges [6]. Our paired construction exposes a residual input difference after exclusion.

Table 5. **Recall at equal realized cost on extracted test faults.** Recall at two realized costs, in untouched compartments per image, obtained by interpolating the common-threshold sweep. Gains are against the released maximum in percentage points. Intervals resample the same original photos across readouts. The scene gate passes every fault in these rows. The development-locked comparison remains the prospective decision test.

Readout	Recall at 0.30	Gain	Recall at 0.50	Gain
Released, max	81.9 [77.5, 85.5]	+0.0 [+0.0, +0.0]	82.6 [79.0, 86.6]	+0.0 [+0.0, +0.0]
Released, 99th pct.	77.5 [73.2, 81.5]	-4.3 [-7.2, -1.8]	77.9 [73.6, 81.9]	-4.7 [-7.6, -2.5]
Footprint, seed 1	99.6 [96.7, 100.0]	+17.8 [+13.4, +22.1]	100.0 [100.0, 100.0]	+17.4 [+13.4, +21.0]
Footprint, seed 2	88.4 [64.9, 96.0]	+6.5 [-16.7, +15.6]	97.1 [94.2, 99.6]	+14.5 [+9.8, +18.8]

Table 6. **Upstream-only continuation on extracted maps.** The original SALAD-style augmentations are used without explicit relocation targets from the pushpin rule. Scene alarms and spatial metrics have the same definitions as the main table.

Map and compartment score	Src. / Sham	Fault / Vac.	Top 2 (%)	Marks to both	Recall (%)	Untouched	Reg. area
Released, max	5 / 7	138 / 138	77.5	2.43	95.7	11.07	81.6
Class-wide, seed 1	15 / 17	138 / 138	71.0	4.83	96.4	12.01	76.1
Footprint, seed 1	18 / 20	138 / 138	76.8	2.56	95.3	7.74	88.1
Class-wide, seed 2	18 / 18	138 / 138	75.4	2.49	96.0	11.50	71.9
Footprint, seed 2	23 / 20	138 / 138	71.7	2.57	95.3	7.07	87.1

Table 7. **Upstream-only continuation on oracle labels.** The released oracle path is shown for comparison.

Map and compartment score	Src. / Sham	Fault / Vac.	Top 2 (%)	Marks to both	Recall (%)	Untouched	Reg. area
Released, max	31 / 31	138 / 138	76.8	2.46	95.3	10.51	81.4
Class-wide, seed 1	16 / 20	138 / 138	60.9	5.02	96.7	12.08	73.7
Footprint, seed 1	21 / 26	138 / 138	77.5	2.50	95.3	7.05	88.1
Class-wide, seed 2	24 / 30	138 / 138	77.5	2.43	97.1	11.75	71.9
Footprint, seed 2	11 / 13	138 / 138	71.0	2.54	97.1	8.04	86.6

Table 8. **Paired differences in the upstream-only continuation.** Untouched compartments per fault and percentage-point differences at pre-specified cutoffs.

Input	Footprint contrast	Untouched	Target recall	Sham alarm	Region area
Extracted	3407 vs. released	-3.33 [-3.90, -2.78]	-0.4 [-1.8, +0.7]	+9.4 [+3.6, +15.9]	+6.6 [+4.5, +8.8]
Extracted	3407 vs. class-wide	-4.28 [-4.93, -3.66]	-1.1 [-2.9, +0.7]	+2.2 [-4.3, +8.7]	+12.1 [+8.8, +15.5]
Extracted	3419 vs. released	-4.01 [-4.65, -3.34]	-0.4 [-2.2, +1.4]	+9.4 [+3.6, +15.9]	+5.5 [+3.5, +7.7]
Extracted	3419 vs. class-wide	-4.43 [-5.14, -3.72]	-0.7 [-2.2, +0.7]	+1.4 [-3.6, +6.5]	+15.1 [+11.7, +18.5]
Oracle	3407 vs. released	-3.46 [-3.99, -2.95]	+0.0 [-1.1, +1.1]	-3.6 [-10.9, +3.6]	+6.8 [+4.6, +9.1]
Oracle	3407 vs. class-wide	-5.03 [-5.67, -4.41]	-1.4 [-2.9, -0.4]	+4.3 [-2.9, +11.6]	+14.4 [+11.0, +18.1]
Oracle	3419 vs. released	-2.46 [-3.00, -1.95]	+1.8 [+0.0, +3.6]	-13.0 [-19.6, -6.5]	+5.2 [+3.1, +7.5]
Oracle	3419 vs. class-wide	-3.71 [-4.39, -3.09]	+0.0 [-1.4, +1.4]	-12.3 [-20.3, -5.1]	+14.7 [+11.6, +17.9]

Table 9. **Upstream-only alternate operating points.** Recalls and region area in percent. Counts are compartments per photo. The released 99th percentile, 95th percentile and reconstruction product results appear in Tab. 4.

Input	Policy	Top 3	98 rec.	98 unt.	Cost rec.	Cost unt.	Fault region	Sham marks	Vac. unt.
Extracted	Released, max	94.2	98.9	12.56	81.9	0.30	74.0	0.00	0.39
Extracted	Class-wide, seed 1	72.5	98.6	12.67	73.9	0.33	73.5	0.00	0.25
Extracted	Footprint, seed 1	88.4	99.6	12.74	89.5	0.74	82.6	0.09	0.48
Extracted	Class-wide, seed 2	92.0	99.6	12.64	70.3	0.21	68.8	0.01	0.46
Extracted	Footprint, seed 2	87.7	99.3	11.28	87.7	0.60	78.5	0.18	1.95
Oracle	Released, max	92.8	98.2	12.32	83.0	0.39	72.7	0.12	0.37
Oracle	Class-wide, seed 1	68.1	97.1	12.34	71.0	0.20	70.9	0.12	0.23
Oracle	Footprint, seed 1	92.8	98.6	12.33	88.4	0.51	82.7	0.17	0.39
Oracle	Class-wide, seed 2	94.9	99.3	12.55	69.9	0.19	68.1	0.12	0.49
Oracle	Footprint, seed 2	89.1	98.9	11.83	88.0	0.46	78.6	0.16	2.14

Table 10. **Development controls and inspection thresholds.** Compartment AUROC is measured on matched development faults. t_s is the scene threshold. t_{95} and t_{98} use 95% and 98% target recall, and t_{budget} uses 0.5 untouched compartments. Numbers are rounded for display. All decisions use full-precision thresholds.

Input	Recipe	Readout	Dev. comp. AUC	t_s	t_{95}	t_{98}	t_{budget}
Extracted	Release	max	97.9	0.17097	0.58810	0.30395	0.94571
Extracted	Release	99th pct.	97.6	0.17097	0.40509	0.17635	0.91538
Extracted	Release	95th pct.	98.2	0.17097	0.24504	0.10780	0.82902
Extracted	Release	reconstruction product	86.6	0.17097	0.40084	0.24098	0.85666
Extracted	Rule-aware	Class-wide, seed 1	48.2	0.23692	0.24065	0.23530	0.99175
Extracted	Rule-aware	Footprint, seed 1	99.6	0.15256	0.22907	0.21939	0.17743
Extracted	Rule-aware	Class-wide, seed 2	48.6	0.16523	0.23033	0.21716	0.99440
Extracted	Rule-aware	Footprint, seed 2	99.2	0.25479	0.24676	0.22138	0.22036
Oracle	Release	max	98.3	0.15802	0.65622	0.38435	0.94875
Oracle	Release	99th pct.	97.7	0.15802	0.48140	0.17617	0.92054
Oracle	Release	95th pct.	97.8	0.15802	0.28629	0.10860	0.82870
Oracle	Release	reconstruction product	90.1	0.15802	0.39195	0.23391	0.86803
Oracle	Rule-aware	Class-wide, seed 1	48.0	0.26617	0.22944	0.21414	0.99222
Oracle	Rule-aware	Footprint, seed 1	99.3	0.14806	0.22603	0.21140	0.19119
Oracle	Rule-aware	Class-wide, seed 2	48.8	0.16464	0.21921	0.19275	0.99473
Oracle	Rule-aware	Footprint, seed 2	98.9	0.27966	0.23471	0.20151	0.21355
Extracted	Upstream-only	Class-wide, seed 1	86.3	0.12738	0.65216	0.45384	0.97235
Extracted	Upstream-only	Footprint, seed 1	97.4	0.15104	0.14474	0.03827	0.39306
Extracted	Upstream-only	Class-wide, seed 2	97.3	0.12330	0.52623	0.29629	0.97085
Extracted	Upstream-only	Footprint, seed 2	97.9	0.13277	0.06335	0.02900	0.24583
Oracle	Upstream-only	Class-wide, seed 1	82.2	0.13064	0.67068	0.56927	0.97446
Oracle	Upstream-only	Footprint, seed 1	97.7	0.15310	0.15520	0.05687	0.41723
Oracle	Upstream-only	Class-wide, seed 2	97.4	0.12206	0.51133	0.29653	0.97182
Oracle	Upstream-only	Footprint, seed 2	97.5	0.17490	0.05590	0.02490	0.26751

Table 11. **Stability on two fixed photo halves.** Difference in untouched compartments per fault, footprint minus class-wide. The 63 and 75 originals were assigned before fitting. The recipe with relocation training is the primary test. The upstream-only continuation reuses the same originals.

Recipe	Input	Seed	Half A (63)	Half B (75)
Rule-aware	Extracted	3407	-12.87 [-12.95, -12.79]	-12.75 [-12.85, -12.63]
Rule-aware	Extracted	3419	-12.63 [-12.76, -12.51]	-12.48 [-12.63, -12.32]
Rule-aware	Oracle	3407	-12.86 [-12.94, -12.76]	-12.75 [-12.84, -12.64]
Rule-aware	Oracle	3419	-12.54 [-12.68, -12.40]	-12.53 [-12.68, -12.37]
Upstream-only	Extracted	3407	-3.98 [-4.92, -3.14]	-4.52 [-5.41, -3.68]
Upstream-only	Extracted	3419	-3.95 [-4.92, -3.02]	-4.84 [-5.81, -3.85]
Upstream-only	Oracle	3407	-4.70 [-5.59, -3.87]	-5.31 [-6.19, -4.44]
Upstream-only	Oracle	3419	-3.10 [-4.05, -2.21]	-4.23 [-5.12, -3.35]

Table 12. **Collateral change after rule-aware adaptation.** Compartments and pixels report percentages with absolute change ≥ 0.5. Max and quantiles are mean within-compartment statistics in sigmoid units. We use the same 13 untouched fault compartments for each variant.

Input	Policy	Edit	Comp. (%)	Pixels (%)	Max	99th pct.	95th pct.
Extracted	Released, max	Fault	89.4	5.4	0.777	0.706	0.578
Extracted	Released, max	Vacancy	2.8	0.5	0.149	0.071	0.039
Extracted	Released, max	Sham	0.0	0.0	0.004	0.001	0.001
Extracted	Class-wide, seed 1	Fault	100.0	6.5	0.980	0.970	0.868
Extracted	Class-wide, seed 1	Vacancy	0.4	0.0	0.056	0.024	0.011
Extracted	Class-wide, seed 1	Sham	0.2	0.0	0.013	0.006	0.003
Extracted	Footprint, seed 1	Fault	0.0	0.0	0.053	0.015	0.005
Extracted	Footprint, seed 1	Vacancy	0.0	0.0	0.017	0.007	0.004
Extracted	Footprint, seed 1	Sham	0.0	0.0	0.002	0.000	0.000
Extracted	Class-wide, seed 2	Fault	100.0	6.5	0.986	0.978	0.885
Extracted	Class-wide, seed 2	Vacancy	0.2	0.0	0.035	0.015	0.006
Extracted	Class-wide, seed 2	Sham	0.0	0.0	0.006	0.002	0.001
Extracted	Footprint, seed 2	Fault	0.0	0.0	0.066	0.027	0.012
Extracted	Footprint, seed 2	Vacancy	0.1	0.0	0.014	0.006	0.002
Extracted	Footprint, seed 2	Sham	0.0	0.0	0.003	0.001	0.001
Oracle	Released, max	Fault	90.3	5.4	0.781	0.711	0.578
Oracle	Released, max	Vacancy	2.9	0.6	0.152	0.075	0.043
Oracle	Released, max	Sham	0.7	0.5	0.010	0.007	0.006
Oracle	Class-wide, seed 1	Fault	100.0	6.9	0.978	0.968	0.861
Oracle	Class-wide, seed 1	Vacancy	0.8	0.5	0.052	0.023	0.012
Oracle	Class-wide, seed 1	Sham	0.7	0.1	0.019	0.010	0.006
Oracle	Footprint, seed 1	Fault	0.1	0.0	0.054	0.016	0.005
Oracle	Footprint, seed 1	Vacancy	0.1	0.0	0.016	0.007	0.004
Oracle	Footprint, seed 1	Sham	0.0	0.0	0.001	0.000	0.000
Oracle	Class-wide, seed 2	Fault	100.0	6.5	0.983	0.975	0.876
Oracle	Class-wide, seed 2	Vacancy	0.9	0.0	0.041	0.020	0.010
Oracle	Class-wide, seed 2	Sham	0.2	0.0	0.010	0.005	0.004
Oracle	Footprint, seed 2	Fault	0.0	0.0	0.067	0.027	0.012
Oracle	Footprint, seed 2	Vacancy	0.1	0.0	0.012	0.005	0.002
Oracle	Footprint, seed 2	Sham	0.0	0.0	0.003	0.001	0.001

Table 13. **Collateral change after upstream-only continuation.** The region definition matches Tab. 12.

Input	Policy	Edit	Comp. (%)	Pixels (%)	Max	99th pct.	95th pct.
Extracted	Released, max	Fault	89.4	5.4	0.777	0.706	0.578
Extracted	Released, max	Vacancy	2.8	0.5	0.149	0.071	0.039
Extracted	Released, max	Sham	0.0	0.0	0.004	0.001	0.001
Extracted	Class-wide, seed 1	Fault	96.5	6.3	0.889	0.848	0.705
Extracted	Class-wide, seed 1	Vacancy	1.8	0.5	0.043	0.020	0.010
Extracted	Class-wide, seed 1	Sham	0.1	0.0	0.008	0.004	0.003
Extracted	Footprint, seed 1	Fault	1.5	0.0	0.171	0.061	0.022
Extracted	Footprint, seed 1	Vacancy	0.9	0.0	0.024	0.009	0.003
Extracted	Footprint, seed 1	Sham	0.0	0.0	0.003	0.001	0.000
Extracted	Class-wide, seed 2	Fault	88.8	5.5	0.816	0.757	0.611
Extracted	Class-wide, seed 2	Vacancy	2.7	0.5	0.100	0.048	0.020
Extracted	Class-wide, seed 2	Sham	0.0	0.0	0.004	0.002	0.001
Extracted	Footprint, seed 2	Fault	1.3	0.0	0.077	0.030	0.009
Extracted	Footprint, seed 2	Vacancy	1.6	0.0	0.036	0.015	0.004
Extracted	Footprint, seed 2	Sham	0.0	0.0	0.001	0.000	0.000
Oracle	Released, max	Fault	90.3	5.4	0.781	0.711	0.578
Oracle	Released, max	Vacancy	2.9	0.6	0.152	0.075	0.043
Oracle	Released, max	Sham	0.7	0.5	0.010	0.007	0.006
Oracle	Class-wide, seed 1	Fault	96.2	6.3	0.892	0.852	0.703
Oracle	Class-wide, seed 1	Vacancy	1.8	0.5	0.042	0.021	0.012
Oracle	Class-wide, seed 1	Sham	0.7	0.5	0.014	0.009	0.008
Oracle	Footprint, seed 1	Fault	1.5	0.0	0.171	0.061	0.023
Oracle	Footprint, seed 1	Vacancy	0.8	0.0	0.022	0.009	0.003
Oracle	Footprint, seed 1	Sham	0.0	0.0	0.002	0.001	0.000
Oracle	Class-wide, seed 2	Fault	90.4	5.5	0.817	0.760	0.608
Oracle	Class-wide, seed 2	Vacancy	2.8	0.6	0.105	0.052	0.023
Oracle	Class-wide, seed 2	Sham	0.7	0.5	0.011	0.008	0.007
Oracle	Footprint, seed 2	Fault	1.4	0.0	0.077	0.030	0.009
Oracle	Footprint, seed 2	Vacancy	1.6	0.0	0.034	0.014	0.004
Oracle	Footprint, seed 2	Sham	0.0	0.0	0.001	0.000	0.000

Table 14. **Released SALAD with all branches on the matched pushpins.** Strict normal-validation cutoffs, separately for each branch and interface. Wins give paired fault-over-sham direction with half ties. The global branch emits a scalar.

Input	Branch	Source	Sham	Fault	Vacancy	New sham	Wins (%)
Oracle	Fusion	77	131	138	138	54	100.0
Oracle	Appearance	9	128	138	137	119	78.3
Oracle	Composition	31	31	138	138	2	100.0
Oracle	Global	82	102	105	119	20	54.3
Extracted	Fusion	75	130	138	138	55	100.0
Extracted	Appearance	9	128	138	137	119	78.3
Extracted	Composition	5	7	138	138	2	100.0
Extracted	Global	84	102	106	119	19	54.3

Table 15. **Response after refitting with mean aggregation.** AUC and top-four recovery use absolute matched probability change and donor-macro aggregation. Top-four change is refit minus original in percentage points. Intervals are donor-paired. The tiers share donors.

Switch tier	Donors	Response AUROC	Top four	Top-four change
Original	12	91.5 [89.1, 93.7]	46.6 [40.5, 52.4]	-40.9 [-49.8, -32.8]
Extra length-matched	11	92.7 [90.1, 95.1]	45.3 [40.2, 50.6]	-37.6 [-48.1, -23.0]
Joint-matched	11	92.9 [90.3, 95.3]	51.4 [41.7, 59.7]	-35.2 [-45.3, -25.6]

6. Graph experiments, diagnostics and native LOCO

The remaining sections describe the graph switch experiments, additional diagnostics of the released composition branch, the native LOCO analysis and an exploratory cardiac query. These are secondary or exploratory analyses with their own protocols.

6.1. Analysis scope

We fixed the GlaS and HVSMR graph folds and the MVTec LOCO partitions before fitting the readouts. The graph protocol froze the graph contrast, support rules, controls, and useful effect sizes before evaluating the new test scores. The pre-specified graph contrast was edited-state loss AUROC minus absolute probability-change AUROC, with a useful positive gap of at least ten points. The measured contrast reverses that direction. A fixed-model degree intervention was prompted by the test outcome and is exploratory. The visual test and the collateral response distribution use all eligible sources. The common-cell comparison of fault, sham, and deletion was a follow-up to the paired spatial output. The screw-bag extent and the native cardiac term probes are secondary follow-ups.

Paired estimands. All graph metrics compare four switch targets with the other pairs within the same graph. The fraction affected and magnitude summaries are computed separately on all targets, added edges, removed edges and untouched pairs. The two-hypothesis AUROC uses half credit for ties. Top-four recall uses fractional allocation to a tied boundary score. Within a source we first average its edits. Within a donor we then average its sources. Donors have equal aggregate weight. The paired bootstrap uses common draw indices across methods and metrics. For image edits, one original photograph and all its variants move together in resampling. Intervals condition on fitted checkpoints and validation cutoffs.

7. Graph construction and matching

GlaS provides 165 gland instance masks from 16 donors. We use the research-use mirror of the original Warwick-QU challenge files, the supplied donor IDs and $0.620\,\mu$m pixel spacing [5]. Graph adjacency is the undirected union of each gland's two closest annotated boundaries. The sensitivity construction takes the undirected union of its two closest centroids. HVSMR-2.0 uses cropped_norm v2 in patient RAS axes. Each nonempty structure defines one typed node. Face contact of at least $5\,\mathrm{mm}^2$ defines a link [4]. Part features include log area or volume, elongation, compactness and channel intensity moments.

Of 165 GlaS fields, 140 have at least four nodes, 129 admit a switch, and 78 meet the original 1.35 length criterion. The 60 cardiac graphs all admit a switch and 59 meet the original bound. The saved manifest includes all original switch pairs. For extra edits we enumerate feasible degree-preserving switches on four distinct nodes and choose up to three distinct hash-ordered edits per source. The additional length tier excludes the saved edit. The alternate construction uses the same length criterion. The joint tier tries both assignments of the two additions to the two removals. It minimizes the maximum absolute log ratio across both raw and locally normalized lengths and requires both attributes to agree within a factor of 1.20. An exact role-pair match is prioritized within this bound. In cardiac graphs, each observed role is unique, so each unordered role pair identifies a single candidate. Matching the role-pair multiset exactly would fix the edge set. Cardiac joint switches are distance-matched. Table 16 reports the coverage tradeoff. The CRAG source paper describes an independent gland cohort [3], but the obtainable tiles lacked a verified patch-to-donor key. All new gland edits and constructions therefore reuse GlaS donors.

An exact topology separation. Place four same-role parts at consecutive vertices of a unit square and two more at $(0.5, -0.8)$ and $(0.5, 1.8)$, using the square from $(0, 0)$ to $(1, 1)$. Join each horizontal side to its adjacent exterior point to make two triangles. Replace the two horizontal links with the two vertical sides. The result is a six-cycle. Degrees are two in both states. All four switched links have raw length one, and all four square vertices have second-nearest centroid radius one. The multiset of roles, link state, raw distance, and local-normalized distance over all pairs is unchanged. A readout of this multiset must tie. Counting triangles or computing shared-neighbor evidence can distinguish the graphs. This construction isolates topology for this restricted representation.

8. Graph scorers and readout controls

Fitting. Four test folds use the original hash-ordered donor allocation. For each family and fold we sort fitting donors by a second salted hash and take the first quarter, with a minimum of two, for development. The rest fit the two learned link models. Both use donor-balanced BCE, with the per-pair weight inversely proportional to the donor's field count and that field's candidate count. AdamW uses weight decay 0.002 and batch size 512. Rates 0.002 and 0.0007 are compared using the smallest weighted development loss. We train up to 80 epochs with patience 12 and initial seeds 904–907. For GlaS there are nine fit and three development donors per fold. For HVSMR there are 34 and 11. Frozen logistic comparators retain all outer fitting donors and use the original 0.015 regularization and feature scales.

Table 16. **Coverage, balance, and sensitivity to matching.** D/F/E gives donors, fields or scans, and edits. Raw and local columns give the median and maximum multiplicative imbalance under the *same* correspondence. The original length criterion uses separately sorted raw lengths, so its maximum under the common correspondence can exceed 1.35. Roles counts exact role-pair matches. Gains are against typed centroid distance. Rank minus response compares the GNN's edited-state loss AUROC with its matched absolute probability-change AUROC. Paired intervals are in points.

Cohort	Tier	D/F/E	Raw	Local	Roles	Local gain	Topol. gain	Rank − response
GlaS	All original	14/129/129	1.18/5.66	1.19/6.71	258/258	+11.8 [+8.9, +15.0]	-2.5 [-5.7, +0.8]	-10.5 [-14.4, -6.2]
GlaS	Centroid 2-NN	11/64/120	1.21/1.87	1.19/1.60	240/240	+15.4 [+11.0, +20.2]	+4.6 [+1.4, +8.1]	-5.1 [-7.1, -2.7]
GlaS	Extra length	11/57/126	1.23/1.46	1.21/1.54	252/252	+16.1 [+10.4, +22.4]	+4.5 [+0.5, +9.2]	-14.5 [-17.4, -11.7]
GlaS	Joint match	11/36/48	1.08/1.20	1.08/1.20	96/96	+16.8 [+11.3, +23.0]	+3.2 [-2.6, +9.6]	-17.1 [-22.8, -12.0]
GlaS	Original length	12/78/78	1.11/1.58	1.10/1.52	156/156	+15.3 [+10.9, +20.1]	+2.1 [-2.4, +7.1]	-16.0 [-19.9, -12.5]
HVSMR	All original	60/60/60	1.08/1.67	1.11/1.83	0/120	+0.4 [-0.2, +1.0]	-5.3 [-6.4, -4.2]	-29.2 [-34.1, -23.8]
HVSMR	Extra length	54/54/146	1.19/1.81	1.17/1.83	0/292	+0.5 [-0.0, +0.9]	-4.4 [-5.6, -3.4]	-27.9 [-30.7, -24.9]
HVSMR	Joint match	43/43/91	1.09/1.19	1.08/1.20	0/182	+0.4 [-0.1, +1.0]	-4.2 [-5.4, -3.0]	-29.9 [-34.8, -24.9]
HVSMR	Original length	59/59/59	1.08/1.67	1.11/1.83	0/118	+0.4 [-0.2, +1.0]	-5.3 [-6.5, -4.1]	-29.5 [-34.5, -24.3]

Table 17. **Graph ranking and response results.** Source and Edited rank targets by the observed-state anomaly score (geometry uses its fixed readout). Response ranks absolute change in conditional probability or change in the geometry readout. The edited intervals are donor paired. GlaS and HVSMR use 78 and 59 source graphs in the original length tier. The pair MLP and frozen GNN controls keep conditional probability fixed.

	GlaS (78 fields, 12 donors)			HVSMR (59 scans, 59 donors)		
Readout	Source	Edited [95% CI]	Response	Source	Edited [95% CI]	Response
Geometry only	64.0	64.0 [57.1, 70.6]	50.0	47.9	47.9 [43.4, 52.6]	50.0
Typed frequency	61.8	61.8 [59.4, 64.2]	50.0	62.4	84.5 [81.9, 86.9]	50.0
Typed distance	70.1	74.9 [70.8, 78.9]	50.0	64.7	87.3 [85.1, 89.5]	50.0
Distance + local spacing	85.9	90.2 [88.4, 91.7]	50.0	65.4	87.7 [85.6, 89.7]	50.0
Handcrafted full context	80.6	85.8 [83.5, 88.0]	99.3	61.2	83.5 [81.3, 85.7]	96.5
Pair MLP	86.7	91.5 [89.1, 93.6]	50.0	68.4	80.4 [77.9, 83.1]	50.0
Masked GNN	77.4	82.5 [78.3, 86.5]	98.5	55.5	59.4 [56.8, 61.8]	88.8
GNN, context frozen	77.4	96.2 [95.0, 97.3]	50.0	55.5	88.5 [85.6, 91.1]	50.0

Frozen references. Typed frequency uses add-one smoothing for the binary outcome within each unordered role pair. The typed centroid logit uses role-pair indicators, the log ratio of distance to the graph median distance, and the squared log-distance term. The log ratio is clipped to $[-5, 3]$. Context adds the log local-spacing ratio, $\log(1 + \mathrm{CN})$, Jaccard overlap, and the logs of one plus the smaller and larger masked degrees. Spacing and topology-only references select the corresponding context features. Feature scales are the fitting standard deviations floored at 0.35. The robust geometry reference uses rolewise medians and 1.4826 times median absolute deviation floored at 0.15. The three shape descriptors of the part reference have a floor of 0.08 and the remaining intensity descriptors a floor of 0.025. The pooled fit is the fallback for unobserved types. All fits and fold assignments are in the reproduction scripts.

Input and capacity. The pair design is the sum of the endpoint one-hot roles, log distance normalized by the median pair distance, and log distance divided by the geometric mean of the two endpoints' local radii. The radii depend only on centroids. Both geometric logs are clipped to $[-5, 3]$ in the learned models. The pair-only MLP has hidden widths 60, 60 and 12 with SiLU activations and 5,065 parameters. For the GNN, we copy the adjacency per query, set both directions of the query to zero, and build the two directed neighborhoods from the masked adjacency. For each neighbor the message has its role, both normalized distances to each endpoint, and a bit indicating incidence to the other endpoint. A $13 \to 32 \to 24$ SiLU network encodes the messages. For each endpoint we sum messages and divide by $\sqrt{1 + \tilde{d}_i}$. We use the componentwise mean and maximum across endpoints plus the sorted $\log(1 + \tilde{d}_i)$ values. The $60 \to 48 \to 24 \to 1$ head includes the same pair inputs, for 5,369 parameters total. No descriptors, adjacency, or embeddings other than the specified pair inputs enter the MLP. GNN development link AUROC spans 97.4–99.1% on the original gland graph and 85.1–91.6% on cardiac. Both models beat the fitting prevalence intercept in all eight original-family folds. The pair MLP meets the 0.80 development AUROC check in all four GlaS and three of four cardiac folds.

Repeatability and held-fixed evidence. We checked an observed link and a non-link on every scored source construction. Reversing the bit for the query alone gives identical inputs and an identical conditional probability in 680/680 checks. Repeated scores on the same input had zero drift.

Reindexing vertices changes the learned probability by at most 4.77×10^{-7}. We use 2×10^{-6} for float32 and 10^{-10} for the original float64 score family, consistent with the original similarity and reindexing controls. Numerical ties are distinct from changes of at least 0.10 development IQR. The IQR is measured on unedited inner development graphs, with a 10^{-3} floor. A very small IQR for the observed-state loss can occur when most source non-links are assigned high confidence.

For the GNN, the two paths through observed-state bit a and logit z give direct term $D = -(a' - a)(z + z')/2$ and context term $C = \Delta s - D$. We report $\sum_T |D|/(\sum_T |D| + \sum_T |C|)$ per edit, then aggregate by source and donor. The mean on GlaS is 65.4% [64.0, 67.2] and on HVSMR 71.1% [68.1, 74.0]. Some changes cancel between terms. The context-frozen model holds z fixed and reads $s(a', z)$. In GlaS its $|\Delta s|$ ranks all targets first, because its observed query bit is the only changing input. A high response of the loss can therefore differ from a high response of the fitted evidence.

Intervening on masked degree. Write the original neighborhood vector as the mean message times $\tilde{d}_i/\sqrt{1 + \tilde{d}_i}$. An empty neighborhood has mean vector zero. The degree-fixed probe uses the edited mean and the source size factor and log degree. The content-fixed probe uses the source mean and the edited factor and log degree. In a degree-preserving switch, a query with unchanged bit has the same masked degree in both states. Consequently the content-fixed intervention changes only the four target queries. These interventions act on fitted network inputs and can create combinations the network did not see in training. Table 19 reports the same protocol on additional edits. Local spacing remains important across constructions, while the learned conditional response has a lower-order cardinality route.

Motion check. A separate perturbation moved one centroid by one median nearest-neighbor spacing while preserving adjacency and all part pixels. Geometry scored 46.8% target AUROC on 140 GlaS fields from 16 donors and 68.9% on 60 cardiac scans. Their paired directional wins were 51.2% and 71.7%, respectively. This intervention answers a different question from an edge switch and can make some configurations more typical.

9. Photograph interventions and extractor

The asset mirror retains the official LOCO directory names and RGB photographs. We freeze 372 train/good, 69 validation/good and 138 test/good source identities and fingerprint each image. Training images are split by hash into 302 fitting and 70 inner development images. The visual holdout divides into 63 and 75 test sources by a separate hash. No test score or activation enters candidate selection. The authors' released maps determine the five-column, three-row tray geometry. On the 256-square grid we use $x = (8, 57, 105, 153, 201, 249)$ and $y = (13, 91, 166, 241)$. Retained pin labels are eight-connected class-2 components of at least 75 pixels. A qualifying source has one per cell. This filter is specific to the compartment experiment.

The RGB cutout is the pin-class component with the neighboring bright pixels from a two-pixel halo, with enclosed holes filled. Selection requires cell clearance and an available local dark background. We fill the original position with the per-channel median of the local dark class-4 ring. Destinations have a one-pixel cell margin, zero overlap with the source cutout or other pins, and at least a two-pixel pin clearance. The oracle component moves entirely onto class 4. The two translations have the same integer squared displacement and equal paste area. The displacement range and intensity thresholds were fixed on validation photographs. In test the moves range from 19.2 to 51.4 pixels. All 138 candidate images meet the edit rules. Sham and fault change exactly the same number of decoded RGB pixels in each pair. Source-pixel repaint is exact. All test masks retain the same class histogram for source, sham and fault. Each move preserves the pin-class component count. All four RGB arms use original camera backgrounds. The held-out scenes contain no generated textures or objects.

The RGB-only extractor is a two-level U-Net with widths 16, 32 and 64, SiLU, group normalization, max pooling and nearest-neighbor upsampling. It predicts six channels. The training label is the released map for each train/good photograph. Weighting is inverse square-root class frequency with floor 0.01. AdamW runs with seed 74817, learning rate 0.001, decay 0.001, batch size 12, and at most 30 epochs, selecting the lowest weighted loss on the 70 inner development images. The selected epoch is 30. Development pin IoU is 97.3% against the released labels and all 70 development images have the correct retained cell counts. On held-out fault maps, pin IoU against the intervened oracle maps is 97.2%. We keep extraction discrepancies in the test denominator. The extracted arm has its own validation cutoff. Its comparison with the oracle arm includes the effect of re-segmentation and recalibration.

Reference regions and scene operation. The two affected compartments are a rule-based coarse spatial reference. The vacant and occupied compartments need no pixel-level anomaly outline. For each cell we take the maximum of either the edited output or the absolute change from source. Even the change-based cell rank is 98.4% on oracle maps while 90.3% of untouched cells have a large absolute change. The matched deletion, which leaves the same vacancy, keeps changes above 0.5 to 2.9% of those same untouched cells. The paired difference is 87.4 points [82.4, 91.9] for oracle

Table 18. **Matched switch sensitivity, direction, and collateral.** T denotes the four targets and U untouched pairs. AUROCs, moved fractions and directional wins are percentages. W_+ and W_- summarize additions and removals. The last five columns use development IQR units and report signed mean target shift and absolute shift percentiles. Each percentile is first taken within an edit. The matched scope has 78 fields from 12 gland donors and 59 scans from 59 cardiac donors. Full paired intervals and raw pair scores are retained in the reproducibility data.

Cohort	Readout	Src.	Edited	$\mathrm{AUC}_{\lvert\Delta s\rvert}$	T moved	U moved	W_+	W_-	Mean Δ_T	T_{50}	T_{90}	U_{50}	U_{90}
GlaS	Part appearance	55.3	55.3	50.0	0.0	0.0	50.0	50.0	+0	0	0	0	0
	Geometry only	64.0	64.0	50.0	0.0	0.0	50.0	50.0	+0	0	0	0	0
	Typed frequency	61.8	61.8	100.0	100.0	0.0	100.0	0.0	+0	881	881	0	0
	Typed distance	70.1	74.9	100.0	100.0	0.0	30.5	79.4	+0.23	2.04	2.68	0	0
	Distance + local spacing	85.9	90.2	100.0	100.0	0.0	41.5	76.9	+0.985	4.58	6.64	0	0
	Distance + topology	70.6	77.0	95.7	100.0	21.6	48.3	65.2	+0.423	3.43	5.61	0.00301	0.517
	Handcrafted full context	80.6	85.8	98.7	100.0	21.6	51.9	70.3	+1.08	5.8	9.76	0.00116	0.309
	Full context + geometry	74.6	80.1	98.7	100.0	21.6	51.9	70.3	+0.369	1.85	3.11	0.0005	0.094
	Pair MLP	86.7	91.5	100.0	100.0	0.0	48.9	71.2	+6.79	29.5	45.7	0	0
	Masked GNN	77.4	82.5	99.3	100.0	54.7	48.1	70.3	+23.3	156	307	0.0264	1.92
	GNN, context frozen	77.4	96.2	100.0	100.0	0.0	73.0	82.3	+290	398	666	0	0
HVSMR	Part appearance	53.1	53.1	50.0	0.0	0.0	50.0	50.0	+0	0	0	0	0
	Geometry only	47.9	47.9	50.0	0.0	0.0	50.0	50.0	+0	0	0	0	0
	Typed frequency	62.4	84.5	100.0	100.0	0.0	75.4	67.8	+1.78	2.71	5.07	0	0
	Typed distance	64.7	87.3	100.0	100.0	0.0	77.1	64.4	+1.45	2.05	3.84	0	0
	Distance + local spacing	65.4	87.7	100.0	100.0	0.0	83.9	63.6	+1.49	2.14	3.94	0	0
	Distance + topology	60.3	82.0	99.6	100.0	28.0	80.5	56.8	+1.18	2.15	3.75	2.88e-05	0.105
	Handcrafted full context	61.2	83.5	99.3	100.0	28.0	81.4	57.6	+1.32	2.3	4	5.78e-05	0.0944
	Full context + geometry	56.0	78.4	99.3	100.0	28.0	81.4	57.6	+0.854	1.48	2.56	3.82e-05	0.0617
	Pair MLP	68.4	80.4	100.0	100.0	0.0	80.5	50.8	+0.848	1.94	3.8	0	0
	Masked GNN	55.5	59.4	85.8	100.0	77.1	76.3	38.1	+0.411	2.16	4.1	0.0874	1.33
	GNN, context frozen	55.5	88.5	100.0	100.0	0.0	92.4	67.8	+4.52	5.94	8.93	0	0

Table 19. **Degree intervention on the fitted GNN.** Response ranks absolute changes in probability. Paired 95% intervals are shown. Repeated edits reuse the original donors.

Cohort	Tier	GNN comparison	Fields/donors	Response AUROC	Top-four recall
GlaS	Original length	Full response	78/12	98.5 [97.6, 99.3]	87.5 [82.4, 92.4]
GlaS	Original length	Degree fixed	78/12	85.3 [81.8, 88.9]	33.7 [25.9, 40.8]
GlaS	Original length	Mean messages fixed	78/12	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
GlaS	Extra length	Full response	57/11	97.6 [95.1, 99.2]	83.0 [70.0, 91.4]
GlaS	Extra length	Degree fixed	57/11	85.4 [80.1, 90.0]	28.3 [21.3, 34.3]
GlaS	Extra length	Mean messages fixed	57/11	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
GlaS	Joint match	Full response	36/11	98.8 [97.9, 99.6]	86.6 [80.1, 93.4]
GlaS	Joint match	Degree fixed	36/11	87.7 [83.4, 91.9]	34.7 [23.4, 46.6]
GlaS	Joint match	Mean messages fixed	36/11	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
HVSMR	Original length	Full response	59/59	88.8 [85.0, 92.5]	67.4 [58.5, 76.3]
HVSMR	Original length	Degree fixed	59/59	74.3 [70.3, 78.3]	41.9 [34.7, 49.2]
HVSMR	Original length	Mean messages fixed	59/59	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
HVSMR	Extra length	Full response	54/54	89.9 [87.5, 92.3]	65.3 [59.5, 70.8]
HVSMR	Extra length	Degree fixed	54/54	71.2 [68.8, 73.6]	34.0 [29.7, 38.1]
HVSMR	Extra length	Mean messages fixed	54/54	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]
HVSMR	Joint match	Full response	43/43	90.3 [86.3, 93.8]	70.4 [61.4, 78.8]
HVSMR	Joint match	Degree fixed	43/43	75.3 [71.6, 79.1]	42.2 [34.7, 50.1]
HVSMR	Joint match	Mean messages fixed	43/43	100.0 [100.0, 100.0]	100.0 [100.0, 100.0]

maps and 86.6 [81.4, 91.2] after extraction. The full threshold curve is Fig. 5 of the main paper. These are changes in the spatial output, whereas scene-level maxima and cutoff decisions are a separate measurement.

The image score is the official maximum sigmoid output. The validation IQRs are 0.00974 (oracle) and 0.01060 (extracted). The raw cutoffs are 0.15802 and 0.17097. The rule comparator sums the absolute deviation from one pin per compartment and adds retained out-of-grid pins. Its cutoff is zero on both paths. The comparator uses the same labels and the same validation rule. On re-extracted maps it yields zero sham alarms (exact 95% upper bound 2.6%), while retaining all 138 fault alarms. Table 20 reports the held-out halves and the paired effect. Figure selection uses the case closest to the median oracle fault-minus-sham score. There were zero missed faults. The second row uses the first hash-ordered alarmed sham, whose source also alarms.

10. Native LOCO and screw-bag analysis

The original SALAD repository checkpoint pair is used for each category. We verified the published download archive, the two checkpoint fingerprints per category, the upstream map decoder, and its unmodified composition forward pass.

Table 20. **Fixed photo halves for the matched photo edits.** Sham, Fault and Del. count alarms. All fault and deletion recalls are 100% on the respective support. Bootstrap intervals for tied all-success contrasts are conditional on the sample.

Maps	Split	Sources	Sham	Fault	Del.	Alarm gap (pp)	Paired wins (%)	Comp. response AUROC
Oracle	A	63	15	63	63	+76.2 [+65.1, +85.7]	100.0 [100.0, 100.0]	99.0 [98.4, 99.6]
Oracle	B	75	16	75	75	+78.7 [+69.3, +88.0]	100.0 [100.0, 100.0]	97.9 [96.3, 99.1]
Oracle	All	138	31	138	138	+77.5 [+70.3, +84.1]	100.0 [100.0, 100.0]	98.4 [97.5, 99.1]
Extracted	A	63	1	63	63	+98.4 [+95.2, +100.0]	100.0 [100.0, 100.0]	99.2 [98.7, 99.6]
Extracted	B	75	6	75	75	+92.0 [+85.3, +97.3]	100.0 [100.0, 100.0]	98.1 [96.4, 99.1]
Extracted	All	138	7	138	138	+94.9 [+90.6, +98.6]	100.0 [100.0, 100.0]	98.6 [97.6, 99.2]

The maps are indexed labels, mapped to six channels without RGB conversion or interpolation. The expected output is a finite 256×256 array in $[0, 1]$. In each category the same validation map repeated exactly. The code normalizes the maximum using its normal validation mean and standard deviation. A positive affine transform preserves all rankings and the corresponding validation-order-statistic threshold. The released inputs contain 1,778 fitting, 305 normal validation and 1,568 test entries. The original benchmark paper reports 1,772, 304 and 1,568. We retain the authors' archive splits and retain the authors' published full-system results as context for the original composition-only analysis [2].

The native matched analysis uses the original component rule with an eight-connected minimum of 12 pixels, a catch-all class unseen in fitting, and both test labels plus at least five fitting maps in the signature. Each signature contributes a positive-weighted AUROC within category. The five-nearest-map baselines use blockwise standardized distances and the same training and calibration maps. Counts are per role. The part bag adds log total area and the median and range of log component areas. The grid adds fractional occupancy per role in a 4×4 grid. The pair block uses the mean coordinate offsets and the 10th, 50th and 90th percentiles of centroid distances by unordered role pair. It uses absolute coordinate offsets within a shared role. Distances are blockwise mean squared standardized differences with training-median imputation. Scale is 1.4826 times the median absolute deviation, floored at 1.0, 0.25, 0.06 and 0.04 for counts, shapes, grid and pairs, respectively. Average the selected block distances before taking their square roots and averaging the five nearest maps. The extended pair variant uses the same counts and morphology. In the spatial block it averages the original pair distance with distances on absolute centroid position and shape. Position uses the 10th, 50th, and 90th percentiles of each coordinate by role. Extent pools log-transformed bounding spans and square roots of the two coordinate covariance eigenvalues at the same quantiles. The robust spread floors for these added blocks are 0.04 and 0.10. Defined coordinate features and fitted-median imputation follow the original procedure.

The matched screw-bag cohort has 79 faults and 95 good maps. Pairs trail the grid by 17.1 points $[-24.9, -9.0]$ from unrounded scores (17.2 from rounded table entries). The other three categories average about -0.3 points. Adding placement and extent gives 56.6% versus 58.3% for the original pair block and 75.5% for the grid. The paired difference relative to the original pair block is -1.7 points $[-9.9, +6.3]$. In Fig. 1, the predicted metal role includes narrow boundary fragments and can merge physical parts. For example, the largest projected long-axis extent in case 121 belongs to a 143-pixel fragment along the image-side edge. Per-role pooled extent therefore cannot identify the actual length of each screw. This probe supports a limit of the tested summaries on the supplied maps. In the native full comparison the grid's logical recall is 51.5% macro with 11.1% good alarms, versus 53.3% and 10.3% for the bag. Matched SALAD minus grid is $+3.0$ points $[-2.1, +8.2]$ across the four supported categories and $+15.1$ [5.9, 24.5] in screw bag.

11. Exploratory SVC–PA query

The code comparison keeps five negative fit patients, 20 negative calibration patients, and test patients with 24 Glenn codes and ten without. Mask face contact has 100% test AUROC, detects 24/24 positives [85.8, 100], and alarms on 1/10 code-negative scans [0.25, 44.5]. The $31.47\,\mathrm{mm}^2$ calibration threshold is below the largest negative (48.45) and smallest positive (58.17). Axial-mask profiles detect 21/24 with zero negative alarms. Full contextual loss has 19.6% AUROC and detects 0/24 at its cutoff. Its masked link probability is higher on the Glenn-coded test scans (median 0.712 versus 0.441). Refitting without the degree terms raises AUROC to 44.2%. Removing the shared-neighbor terms gives 18.3%. Distance alone gives 75.4%. Removing one incident link at a time while holding the SVC–PA bit fixed yields mean absolute within-patient probability shifts of 0.093 on the 21 Glenn-coded patients with at least one such link and 0.144 on ten code-negative patients. These are exploratory sensitivity checks on five fit patients and a single query. They do not resolve the causal effects of surgery. Five target-negative SVC variants occur in the release, with two in fit and three in calibration. The test has zero. Independent adjudication of native relations and benign variants is needed to extend the scope beyond this code.

Table 21. **Native map scores and composition branch alarms.** AUROC in percent. Pos/good counts positives and their eligible negative comparison images. Cutoffs come from normal validation only and stay fixed across scopes. Macro AUROC weights categories equally. Macro alarm counts add images. Pushpins has no eligible matched signature.

Category	Scope	Pos/good	Bag	Grid	Pairs	Extent	SALAD [95% CI]	Hits	Good alarms
Breakfast box	Logical	83/102	79.8	79.9	76.6	78.7	79.0 [71.6, 85.8]	53/83	15/102
	Matched	42/77	71.9	68.3	68.1	71.5	64.4 [53.4, 75.3]	17/42	14/77
	Structural	90/102	68.6	69.1	69.7	68.0	75.4 [68.2, 82.1]	48/90	15/102
Juice bottle	Logical	142/94	84.3	85.6	85.3	83.2	88.1 [83.5, 92.1]	105/142	4/94
	Matched	99/91	82.0	84.1	85.3	82.6	82.9 [77.0, 88.4]	64/99	2/91
	Structural	94/94	74.1	70.9	68.8	66.3	85.3 [79.5, 90.6]	59/94	4/94
Pushpins	Logical	91/138	77.2	76.9	73.3	67.1	99.7 [99.2, 100.0]	91/91	31/138
	Structural	81/138	56.1	59.8	53.7	50.0	76.4 [69.4, 82.9]	47/81	31/138
Screw bag	Logical	137/122	76.0	76.8	63.3	61.6	90.8 [86.6, 94.4]	111/137	14/122
	Matched	79/95	75.7	75.5	58.3	56.6	90.5 [85.1, 95.4]	65/79	9/95
	Structural	82/122	71.8	76.8	67.4	66.4	85.1 [79.3, 90.2]	50/82	14/122
Connectors	Logical	108/119	80.9	81.3	76.6	77.8	82.7 [76.8, 88.1]	74/108	9/119
	Matched	30/70	70.8	80.2	78.2	83.9	82.4 [74.6, 89.0]	21/30	6/70
	Structural	85/119	72.6	72.1	71.6	70.2	92.0 [87.8, 95.7]	69/85	9/119
Macro	Logical	561/575	79.6	80.1	75.0	73.7	88.1 [85.8, 90.2]	434/561	73/575
	Matched	250/333	75.1	77.0	72.5	73.7	80.0 [76.2, 83.8]	167/250	31/333
	Structural	432/575	68.6	69.7	66.2	64.2	82.8 [80.1, 85.4]	273/432	73/575

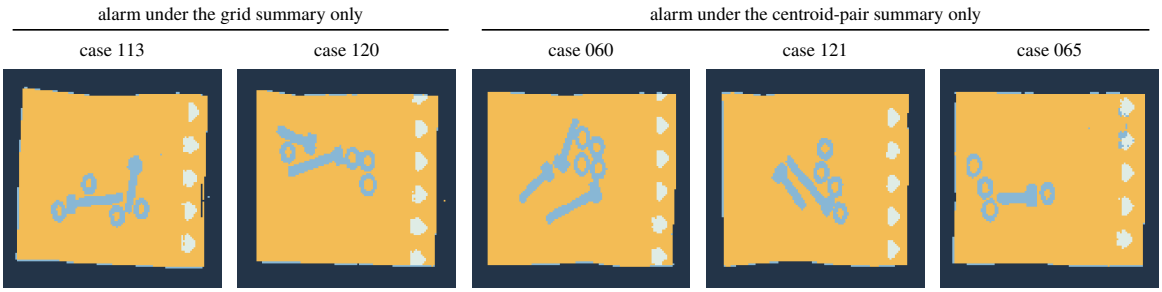


Figure 1. **Screw-bag logical cases with discordant grid and centroid-pair alarms.** At the pre-specified thresholds, two of 79 supported faults alarm under the grid alone and three under the pair summary alone. The colorized maps show the five cases. Thin fragments around the bag and merged foreground illustrate the difference between predicted counts and physical screw inventory.

12. Data use

The medical experiments use existing de-identified research releases and their original labels, and constitute a secondary analysis without new ethics review, participant recruitment, intervention or consent collection. The HVSMR authors report approval by the Boston Children's Hospital IRB with a waiver of written consent [4]. The photographs derive from the public MVTec LOCO AD dataset, distributed under CC BY-NC-SA 4.0, and derived image panels retain this attribution and license. Selected pairs, data splits, scores, calibration sets, checkpoint hashes and figure inputs are released with the code.

LLM Usage Statement

OpenAI Codex was used to develop and implement the revision experiments, run and check the code, inspect public sources and data provenance, produce figures and tables from recorded outputs, and draft and edit the main text and supplementary material. Anthropic Claude was used to restructure and edit the text and to redraw the figures from the recorded outputs. Photograph edits are deterministic composites of the stated dataset assets.

References

[1] Paul Bergmann, Kilian Batzner, Michael Fauser, David Sattlegger, and Carsten Steger. Beyond dents and scratches: Logical constraints in unsupervised anomaly detection and localization. *International Journal of Computer Vision*, 130(4):947–969, 2022. 1, 2

[2] Matic Fučka, Vitjan Zavrtanik, and Danijel Skočaj. SALAD – semantics-aware logical anomaly detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 21843–21852, 2025. 1, 13

[3] Simon Graham, Hao Chen, Jevgenij Gamper, Qi Dou, Pheng-Ann Heng, David Snead, Yee Wah Tsang, and Nasir Rajpoot. MILD-Net: Minimal information loss dilated network for gland instance segmentation in colon histology images. *Medical Image Analysis*, 52:199–211, 2019. 9

[4] Danielle F. Pace, Hannah T. M. Contreras, Jennifer Romanowicz, Shruti Ghelani, Imon Rahaman, Yue Zhang, Patricia Gao, Mohammad Imrul Jubair, Tom Yeh, Polina Golland, Tal Geva, Sunil Ghelani, Andrew J. Powell, and Mehdi Hedjazi Moghari. HVSMR-2.0: A 3D cardiovascular MR dataset for whole-heart segmentation in congenital heart disease. *Scientific Data*, 11: 721, 2024. 9, 14

[5] Korsuk Sirinukunwattana, Josien P. W. Pluim, Hao Chen, Xiaojuan Qi, Pheng-Ann Heng, Yun Bo Guo, Li Yang Wang, Bogdan J. Matuszewski, Elia Bruni, Urko Sanchez, Anton Böhm, Olaf Ronneberger, Bassem Ben Cheikh, Daniel Racoceanu, Philipp Kainz, Michael Pfeiffer, Martin Urschler, David R. J. Snead, and Nasir M. Rajpoot. Gland segmentation in colon histology images: The GlaS challenge contest. *Medical Image Analysis*, 35:489–502, 2017. 9

[6] Jing Zhu, Yuhang Zhou, Vassilis N. Ioannidis, Shengyi Qian, Wei Ai, Xiang Song, and Danai Koutra. Pitfalls in link prediction with graph neural networks: Understanding the impact of target-link inclusion and better practices. In *ACM International Conference on Web Search and Data Mining*, pages 994–1002, 2024. 4